

Inferring Cytokine Signaling Cascade *Direction* from a Single-Time-Point Single-Cell Snapshot

A multiple-instance-learning pipeline and the `cross_asym` fix

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2026-05-31

Abstract

Question. Cytokines are signaling proteins immune cells use to talk to one another. A *cascade* $X \rightarrow Y$ means stimulus X causes cells to produce Y . Cascade *direction* is a temporal, causal property — yet single-cell experiments usually give us only *one* snapshot in time. Can we recover direction from a single frame? **Method.** We build a three-step pipeline on pseudo-bulk “pseudo-tubes” of single cells: (A) an attention-based multiple-instance-learning (AB-MIL) model finds *coupled* cytokine pairs; (B) per-cytokine binary classifiers + Integrated Gradients discover each cytokine’s gene signature S_X ; (C) a directional-asymmetry test assigns direction. **Result.** The textbook §24 score we started from turns out to be *algebraically symmetric* and therefore direction-blind. Replacing it with the *antisymmetric* cross-engagement `cross_asym` $= s(a, S_b) - s(b, S_a)$ recovers direction at **88%** on human PBMC (24h, 17 audited cascades; vs. 47% = chance for the old score) and **86%** on mouse BMDM at 5h (single-frame, crystal-clear TLR-biology labels). The fix is exactly the intuition “the upstream tube carries *both* signatures; the downstream tube carries only its own”.

1 Project map (one page to orient)

This project asks two linked questions about cytokine biology, and they have become two “paths” (Fig. 1). **Path A — axis discovery** (largely complete) asks *which cytokines are coupled*: it recovered 121 cytokine coupling “axes” on a 91-cytokine human dataset, about half supported by literature versus $\sim 1\%$ by chance. **Path B — cascade direction** (this report) asks the harder question: *given a coupled pair, who is upstream?* Path B stalled for months because every attempt scored at chance. *This session found why and fixed it*: the scoring statistic was symmetric and could not encode direction; the antisymmetric `cross_asym` does. That single change is the spine of this talk.

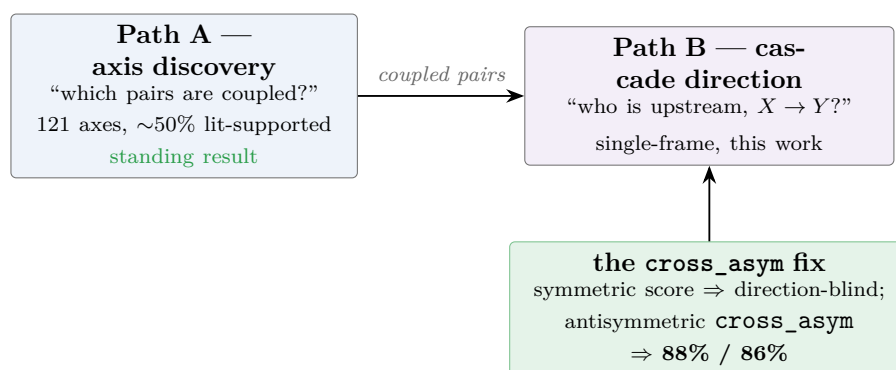


Figure 1: The two paths. Path A (coupling) feeds Path B (direction). This report is Path B; the green node is the contribution.

What this report covers: the biology and why direction-from-a-snapshot is hard but valuable (§2); the data and the model (§3, §4); the result and an honest accounting of its limits (§5–§6);

and a suggested 45-minute talk structure (§8). Exact hyperparameters and equations are in the appendices.

2 Motivation: the biological question

2.1 A five-minute biology primer

Every cell carries the same DNA but *expresses* a different subset of genes; the set of currently-active genes (the *transcriptome*) is the cell’s state. **Cytokines** are small secreted proteins — IFN- β , TNF, the interleukins (IL-x) — that immune cells use to coordinate a response. When a cytokine X binds its receptor on a target cell, it triggers an intracellular *signaling pathway* (a relay of proteins) that switches on a characteristic set of genes, X ’s *signature* S_X .

A **cascade** is when one cytokine begets another (Fig. 2): X stimulates a cell, the cell’s program includes *producing and secreting* cytokine Y , and Y then drives *its* program in neighbours. Cascades are the wiring diagram of the immune system. Knowing the *direction* of an edge ($X \rightarrow Y$ vs. $Y \rightarrow X$) is knowing causality — which is what you need to choose a drug target.

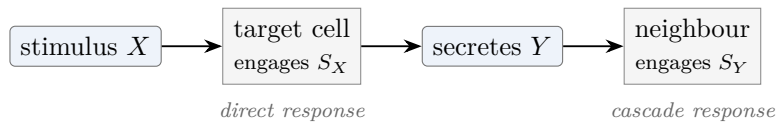


Figure 2: A cascade $X \rightarrow Y$: X ’s target cells run S_X and make Y ; Y then runs S_Y in other cells. The asymmetry we exploit: the X -tube ends up carrying both S_X and S_Y , the Y -tube only S_Y .

2.2 The question and why it is hard

We measure cells with single-cell RNA sequencing, which gives a *snapshot*: the transcriptome of thousands of cells at *one* moment after stimulation (Fig. 3). Direction is a property of *time* — X rises before Y — so the naive view is that a single frame cannot reveal it. A full time-course (sample every hour for many hours, every condition, replicated) would make direction trivial, but it is expensive, and many valuable datasets are single-frame. **The bet of this project: a single frame already contains a *spatial-in-gene-space* fingerprint of direction**, because the upstream stimulus’s cells carry the downstream signature too, while the converse does not.

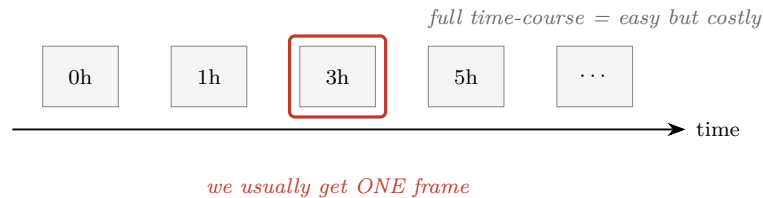


Figure 3: The challenge. A time-course makes direction trivial; a single snapshot (red) discards the time axis. We ask whether direction survives in one frame.

Why it matters. (i) Direction = causality = candidate intervention points. (ii) It unlocks the many single-time-point datasets already deposited. (iii) The method is generic: any system with labeled stimuli that participate in cascades (not just cytokines) could be analysed the same way.

3 Data: cells, pseudo-tubes, and two datasets

3.1 From cells to “pseudo-tubes”

Single-cell RNA-seq yields a (cells $\times$ genes) matrix per condition. A single cell is noisy, and our label (“which stimulus was applied”) is defined at the *sample* level, not the cell level. We therefore build **pseudo-tubes**: a pseudo-tube is a *bag* of cells (stratified across cell types) drawn from one (donor, stimulus) sample (Fig. 4). This matches the multiple-instance-learning setting (§4.2) — a bag has a label, its instances (cells) do not.

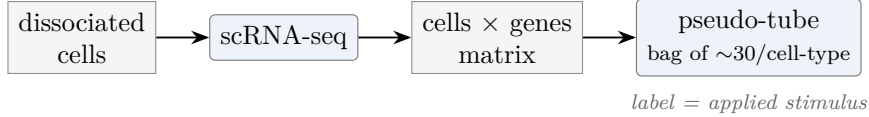


Figure 4: Pre-processing. Cells are sequenced, normalised (total-count + log1p), then assembled into stratified pseudo-tubes; the tube inherits the stimulus label.

3.2 Two complementary datasets

We use two datasets with opposite strengths (Table 1). Oesinghaus (human PBMC, 91 cytokines, 24h) is broad but single-time. Sheu (mouse bone-marrow macrophages, 7 stimuli, a true time-course) is narrow but has the *time axis* and textbook-clean cascade biology — ideal for validating direction.

Table 1: The two datasets.

	Oesinghaus 2024	Sheu 2024
organism / tissue	human PBMC	mouse BMDM (macrophages)
stimuli	91 cytokines (+PBS)	7 (LPS, LPSlo, P3CSK, PIC, TNF, CpG, IFN- β)
time points	24h (single)	0/0.25/0.5/1/3/5/8/24h (course)
genes	4000 HVGs	500-gene targeted panel
“donors”	12 biological donors	4 pseudo-donors @3h (context $\times$ rep)
cells/tube	~ 480 (16 cell types)	~ 120 (mac_c0-c3)
role here	broad benchmark (Path B)	single-frame, multi-time confirmation

3.3 The test cascades (textbook TLR biology)

Sheu’s stimuli are Toll-like-receptor (TLR) agonists with *known* wiring (Fig. 5). Two adaptor arms matter: **TRIF** (engaged by TLR3 and TLR4) $\rightarrow$ IRF3 $\rightarrow$ type-I interferon (IFN- β); and **MyD88** (TLR2/4/9) $\rightarrow$ NF- κ B $\rightarrow$ TNF. So *polyIC* (TLR3) and *LPS* (TLR4) drive IFN- β (an IFN cascade), while *LPS/LPSlo/P3CSK/CpG* drive autocrine TNF (an NF- κ B cascade). TLR2 (P3CSK) has *no* TRIF arm, so P3CSK $\rightarrow$ IFN- β is a true negative. These give us crystal-clear directional labels (Appendix B).

4 Method: the three-step pipeline

4.1 Overview

The pipeline (Fig. 6) chains three modules. *Path A* uses a trained AB-MIL model’s latent geometry to propose coupled pairs. The *Bridge* trains one binary classifier per cytokine and

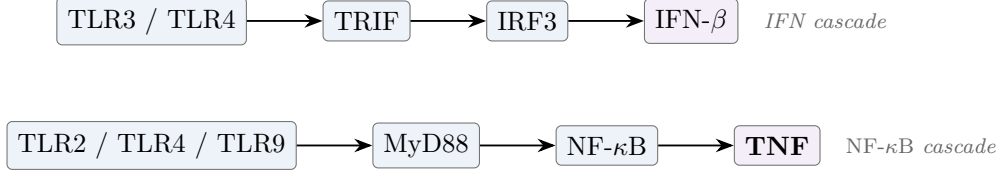


Figure 5: The two adaptor arms. polyIC/LPS $\rightarrow$ IFN- β (TRIF $\rightarrow$ IRF3); the MyD88 ligands $\rightarrow$ autocrine TNF. The two product signatures (ISGs vs. NF- κ B targets) are transcriptionally distinct — the precondition our test needs.

uses Integrated Gradients to read out each cytokine’s gene signature S_X . *Path B* combines the signatures with the cells to score direction. Path A and the Bridge both rest on the same AB-MIL model, described next.

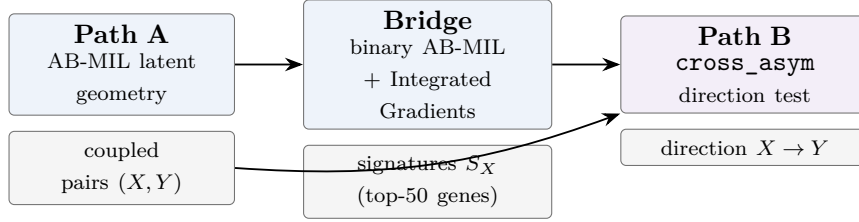


Figure 6: The pipeline. Path A proposes *which* pairs couple; the Bridge discovers per-cytokine signatures; Path B assigns *direction* to the coupled pairs.

4.2 The AB-MIL model

A pseudo-tube is a bag $X \in \mathbb{R}^{N \times G}$ of N cells over G genes, with one tube-level label. Attention-based multiple-instance learning (Fig. 7) maps each cell to an embedding, learns a soft attention weight per cell, pools into one tube vector, and classifies:

$$h_i = \text{InstanceEncoder}(x_i) \in \mathbb{R}^d, \quad a_i = \frac{\exp(w^\top \tanh(Vh_i))}{\sum_j \exp(w^\top \tanh(Vh_j))}, \quad (1)$$

$$z = \sum_i a_i h_i, \quad \hat{y} = \text{softmax}(Wz + b). \quad (2)$$

The encoder is a residual MLP (LayerNorm $\rightarrow$ Linear $\rightarrow$ GELU blocks, skip connections, *no dropout* so attention is stable); the attention is the gated form of Ilse et al. (2018). Training is two-stage: Stage 1 pre-trains the encoder to classify cell types; Stage 2 freezes the encoder and trains attention + head on the tube label. Exact dimensions and schedules are in Appendix A.

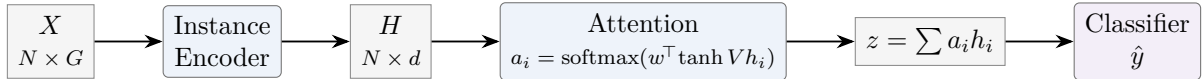


Figure 7: AB-MIL forward pass. Cells $\rightarrow$ embeddings $\rightarrow$ attention pooling $\rightarrow$ tube vector $\rightarrow$ stimulus prediction.

4.3 Path A — coupled pairs (brief)

After Stage 2, we read the encoder’s latent geometry: per cell type, how far each cytokine’s centroid is pushed toward another cytokine’s, relative to the resting (PBS) state. Pairs with significant mutual bias are “coupled axes”. On Oesinghaus this produced 121 axes ($\sim 50\%$ literature-supported). For this report Path A is context; the spotlight is the Bridge and Path B.

4.4 Bridge — discovering signatures with Integrated Gradients

For each cytokine X we train a *binary* AB-MIL classifier (X vs. PBS). **Integrated Gradients** (IG) then attributes the prediction to input genes: it integrates the model’s gradient along a straight line from a baseline tube (the PBS per-gene mean) to the real tube (20-step midpoint rule), giving a per-gene importance. Averaging over tubes and ranking yields S_X = the top-50 genes. Crucially S_X is *discovered from data*, not curated by hand.

4.5 Path B — direction, and the bug we fixed

Given a coupled pair (a, b) with signatures S_a, S_b , let $s(x, S)$ be the mean expression of gene-set S in x ’s tubes (PBS-normalised). The original “§24” directional score was

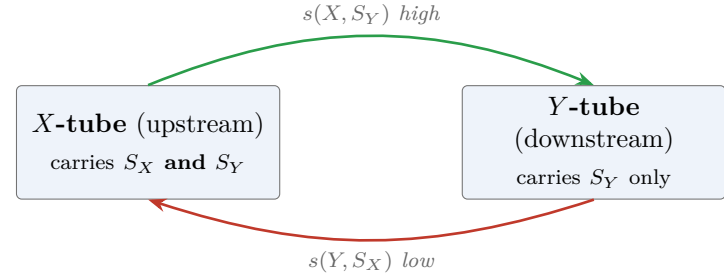
$$\text{directional_score} = \underbrace{(s(a, S_a) - s(b, S_a))}_{\text{asym on } S_a} - \underbrace{(s(a, S_b) - s(b, S_b))}_{\text{asym on } S_b} = (s_a S_a + s_b S_b) - (s_a S_b + s_b S_a). \quad (3)$$

Expanding shows it is **symmetric under swapping** $a \leftrightarrow b$: its *sign cannot encode direction* when the two “pathways” are the cytokines’ own signatures. This is exactly why our first runs scored at chance.

The direction lives in the **antisymmetric cross-engagement**:

$$\boxed{\text{cross_asym}(a, b) = s(a, S_b) - s(b, S_a)} \quad (\text{PBS-normalised}). \quad (4)$$

It flips sign under $a \leftrightarrow b$. Its meaning is the cascade intuition (Fig. 8): if a is upstream, a ’s tubes carry b ’s signature (via the cascade) so $s(a, S_b)$ is high, while b ’s tubes do *not* carry a ’s signature so $s(b, S_a)$ is low $\Rightarrow \text{cross_asym} > 0 \Rightarrow a$ upstream. We aggregate the per-cell-type **cross_asym** by its median and report a sign-consensus fraction; a permutation null with random gene-sets guards against the signatures merely tracking activation level.



$$\text{cross_asym}(X, Y) = s(X, S_Y) - s(Y, S_X) > 0 \Rightarrow X \text{ upstream}$$

Figure 8: Why **cross_asym** encodes direction. The upstream tube engages the downstream signature more than vice versa; **directional_score** (symmetric) averages this away, **cross_asym** (antisymmetric) keeps it.

5 Results

5.1 Oesinghaus 24h: 88% vs. 47%, same data

We first hand-audited the human-PBMC literature labels (the automatic keyword tags were noisy; 7 of 29 flipped under receptor-biology review), giving 17 cleanly directional benchmark cascades. On *identical* per-cell-type values, swapping **directional_score** for **cross_asym** lifts directional sign-accuracy from $8/17 = 47\%$ (chance) to $15/17 = 88\%$ (Fig. 9, Table 2). The two misses

both involve VEGF. A random-gene-set permutation null is cleared by 34 axes ($p < 0.05$), and a label-permutation null on the benchmark gives $p = 0.003$ — the discovered signatures carry genuine direction information, not just activation magnitude. The per-axis breakdown (Fig. 5.1) shows the sign is right across diverse cascades (interferon, interleukin, growth-factor).

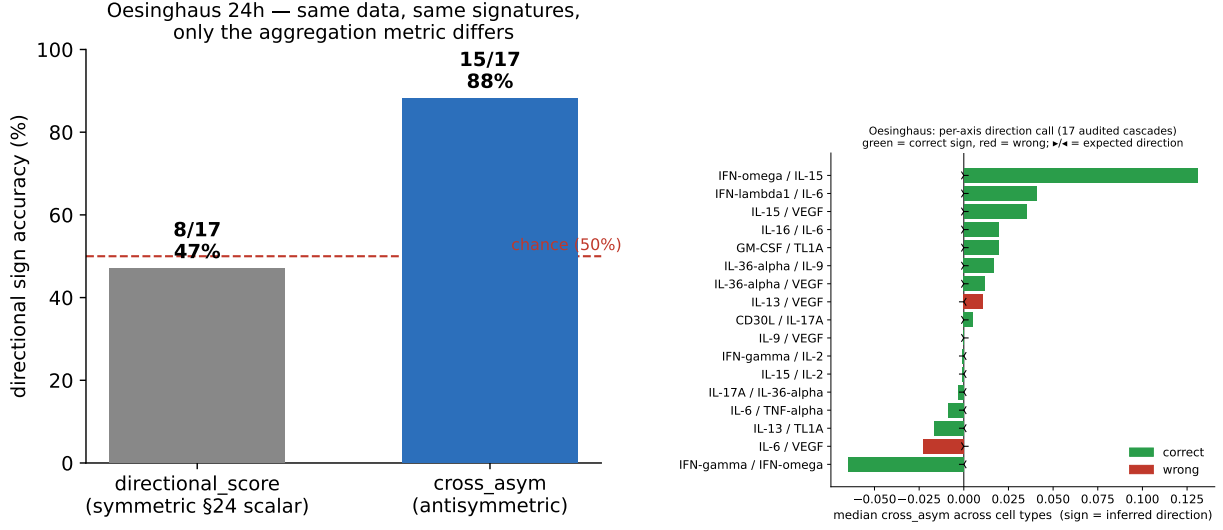


Figure 9: Left (F9): the headline — the only change is the aggregation metric. **Right (F10):** per-axis median `cross_asym` on the 17 audited cascades; green = correct sign, red = wrong (both VEGF). ▶ / ◀ mark the expected direction.

Table 2: Oesinghaus 24h, 17 audited directional cascades.

metric	a→b (n=10)	b→a (n=7)	total
directional_score (symmetric)	—	—	8/17 = 47%
cross_asym (antisymmetric)	9/10 = 90%	6/7 = 86%	15/17 = 88%

34/53 axes beat the random-gene-set null ($p < 0.05$); benchmark label-permutation $p = 0.003$.

5.2 Sheu single-frame, multiple time points

Each time point is analysed *independently* — no information from one frame enters another (the core single-frame constraint). On crystal-clear TLR-biology labels, directional accuracy is best at **5h**: **6/7 = 86%** (Table 3, Fig. 11). Two findings stand out (Fig. 10): (i) the NF- κ B→TNF cascades recover *well* (4/4 at 5h) — the opposite of the older curated-pathway approach, because the data-discovered signatures separate TNF from the TLR ligands; (ii) one IFN cascade, **polyIC**→IFN- β , fails at *every* time point.

Table 3: Sheu single-frame directional accuracy by class × time.

class	1h	3h	5h
IFN (MUST: LPS, polyIC →IFN- β)	1/2	1/2	1/2
IFN (SHOULD: LPSlo→IFN- β)	—	1/1	1/1
NF- κ B (SHOULD: · →TNF)	3/3	3/4	4/4
all directional	4/5	5/7	6/7

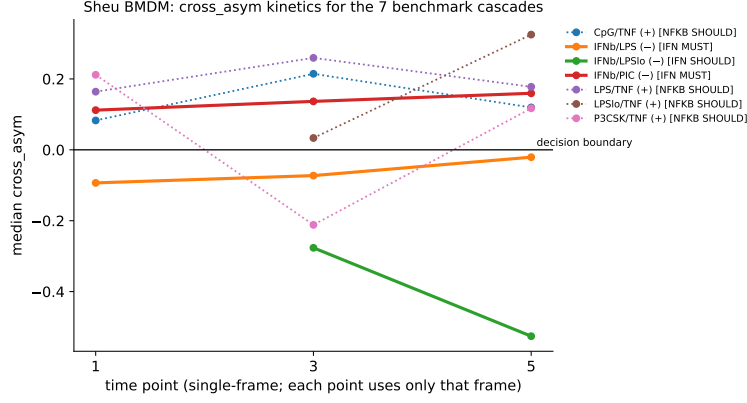


Figure 10: (F11) `cross_asym` kinetics for the 7 Sheu benchmark cascades; 0 is the decision boundary. IFN cascades solid, NF- κ B dotted. LPS/IFN- β sits correctly below 0 at all times; polyIC/IFN- β sits wrongly above.

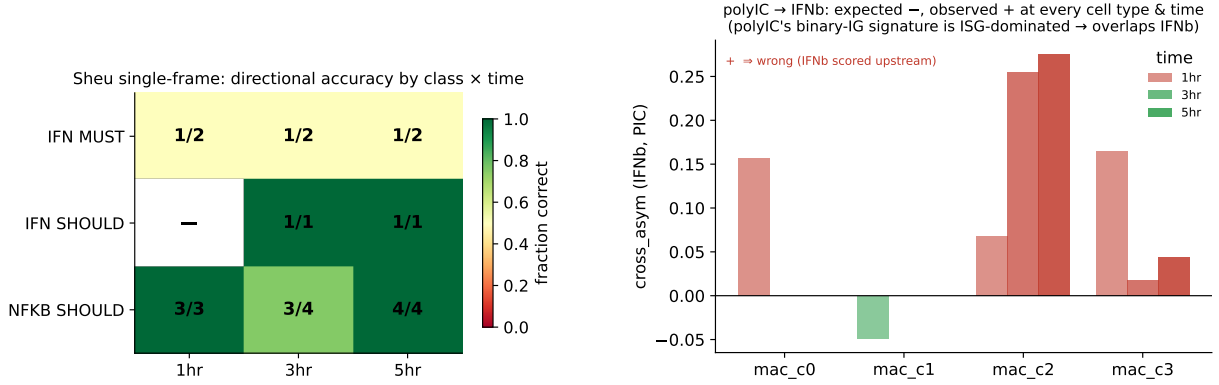


Figure 11: Left (F12): accuracy heatmap (class $\times$ time); 5h is best. Right (F13): the polyIC $\rightarrow$ IFN- β failure is consistent across all cell types and times (all positive = wrong direction).

5.3 Why polyIC $\rightarrow$ IFN- β fails (and why that is informative)

polyIC's *discovered* signature is dominated by interferon-stimulated genes (ISGs): those are the strongest differentially-expressed genes, far above the lower-expressed IRF3-direct genes (*Ifnb1*, *Cxcl10*). So $S_{\text{polyIC}} \approx S_{\text{IFN-}\beta}$, and IFN- β (which signals *directly* through its receptor) engages this shared ISG set more strongly than polyIC (which reaches it only via the autocrine loop). The sign therefore inverts. LPS $\rightarrow$ IFN- β survives because LPS carries a broad non-ISG (NF- κ B) program that keeps S_{LPS} distinct from $S_{\text{IFN-}\beta}$. This is a concrete, testable failure mode (next steps, §7), not a mystery.

6 Honest limitations

- **Direction, not existence.** `cross_asym` assigns a direction; it does *not* decide whether a cascade exists. No-cascade Sheu pairs also have large $|\text{cross_asym}|$. That is by design: Path A decides *which* pairs couple; `cross_asym` only directs those.
- **Small benchmarks.** 17 and 7 graded cascades. Sheu's IFN-MUST is effectively one clean win (LPS) plus one explained failure (polyIC).
- **Labels are a judgement call.** The 88% is against *our* receptor-biology audit (7 tags flipped); the audit log is open for review.

- **Retrospective.** We test on *known* cascades; we have not yet made a prospective, experimentally-validated prediction.

7 Current state and next steps

The method works at $\sim 86\text{--}88\%$ direction accuracy on two independent datasets, including early time points, with a clean mechanistic story. Cheap next experiments: (1) **polyIC ISG-exclusion test** — re-derive S_{polyIC} with ISGs removed; we predict the sign flips to correct, isolating the mechanism; (2) inspect the VEGF signature (the two Oesinghaus misses); (3) run Path A $\rightarrow$ Path B *end-to-end* on Sheu (coupling then direction); (4) sensitivity to the top- n signature size. The big-picture goal is a prospective prediction confirmed by a perturbation experiment.

8 Suggested 45-minute talk structure

Table 4: A 45+15 plan. Anchor on the four “money” figures (F2, F8, F9, F12).

min	segment	key visual
0–8	biology primer + the question (why direction is hard/important)	F2, F3
8–16	data + test-cascade biology (TLR arms)	Table 1, F5
16–31	method: MIL $\rightarrow$ IG signatures $\rightarrow$ cross_asym (the symmetric-bug story)	F6, F7, F8
31–41	results: Oesinghaus 88%, Sheu kinetics, the polyIC wart	F9 , F11, F12
41–45	limitations + next steps	—
45–60	discussion	—

A Full technical specifications

Table 5: Model architecture (dimensions) across runs.

run	embed d	enc. h_0	enc. h_1	att. hidden	classes
Sheu Stage 1+2 (Path A)	32	512	256	16	8
Sheu binary (Bridge)	128	256	128	64	2
Oesinghaus binary (narrow, 8)	32	128	64	16	2
Oesinghaus binary (wide, 16)	512	512	512	128	2

Encoder: input_proj(LayerNorm, GELU) $\rightarrow$ ResBlock(h_0) $\rightarrow$ DownBlock($h_0 \rightarrow h_1$) $\rightarrow$ ResBlock(h_1) $\rightarrow$ DownBlock($h_1 \rightarrow d$); GELU; Kaiming init; block fc2 zero-init (identity start); **no dropout**. Attention: gated Ilse-2018, $V \in \mathbb{R}^{d \times \text{att}}$, w linear, Xavier init. Classifier: Linear(d, K).

Table 6: Training schedules (SGD, momentum 0.9).

run	S1 epochs	S1 lr	S2 epochs	S2 lr	warmup
Sheu Stage 1+2	30	0.003	40	0.0005	5
Sheu binary	20	0.005	200	0.0001	—
Oesinghaus binary (both)	20	0.005	250	3e-5	—

Stage 1 = encoder pre-train on cell types (batch 256); Stage 2 = frozen-encoder MIL on tube labels (one tube per step). Sheu val pseudo-donor M2_IL4_rep1; Oesinghaus val donors 2,3.

Table 7: Bridge (IG), Path B, and pseudo-tube parameters.

Integrated Gradients	baseline = PBS per-gene mean; 20-step midpoint rule; ≤ 10 tubes/cytokine; S_X = top-50 by IG rank; CPU
Path B	top_n 50; min_cells 10; n_null_perms 100; magnitude_threshold 0.01; strong/weak consensus 0.75/0.50
pseudo-tubes	30 cells/cell-type; min 10 cells; 10 tubes/(stim,donor); cell-typing by global Leiden on resting cells

Key equations. Encoder/attention/classifier as in §4.2. Integrated Gradients for gene g : $IG_g = (x_g - \bar{x}_g^{\text{PBS}}) \cdot \frac{1}{n} \sum_{k=1}^n \partial_{x_g} f(\bar{x}^{\text{PBS}} + \alpha_k(x - \bar{x}^{\text{PBS}}))$, $\alpha_k = \frac{k-0.5}{n}$. Direction: $\text{cross_asym}(a, b) = (s(a, S_b) - s_{\text{PBS}}(S_b)) - (s(b, S_a) - s_{\text{PBS}}(S_a))$, $\text{sign} > 0 \Rightarrow a$ upstream.

B The 21 Sheu pairs (crystal-clear labels)

Table 8: Hand-curated from TLR receptor biology. Benchmark = the 7 directional cascades scored; negatives test specificity; unknown = excluded.

pair	class	reason
polyIC / IFN- β	IFN MUST	TLR3→TRIF→IRF3→IFN- β ; cleanest cascade
LPS / IFN- β	IFN MUST	LPS engages the TLR4–TRIF arm
LPSlo / IFN- β	IFN SHOULD	low-dose LPS retains TRIF arm (weaker)
LPS / TNF	NF- κ B SHOULD	TLR4→NF- κ B→autocrine TNF
LPSlo / TNF	NF- κ B SHOULD	low-dose TLR4→NF- κ B→TNF
P3CSK / TNF	NF- κ B SHOULD	TLR2→NF- κ B→TNF
CpG / TNF	NF- κ B SHOULD	TLR9→NF- κ B→TNF
P3CSK / IFN- β	NEGATIVE	TLR2 has no TRIF arm $\Rightarrow$ no IFN- β
CpG / IFN- β	NEGATIVE	TLR9 type-I IFN is pDC-restricted; minimal in BMDM
TNF / IFN- β	NEGATIVE	no TNF→IFN- β cross-induction in macrophages
LPS / LPSlo	NEGATIVE	same receptor, dose difference — not a cascade
<i>Plus 10 cross-receptor / MyD88-overlap pairs (CpG/LPS, LPS/polyIC, ...) = UN-</i>		
KNOWN (excluded from scoring).		

Reproducibility: figures from `scripts/make_report_figures.py`; results from `run_pipeline_a_bridge_b(_sheu).py` + `retally_pipeline_against_audit.py`; labels in `reports/sheu_cascade/` and `reports/cascade_pairs/`.

Full session narrative: `reports/SESSION_SUMMARY_2026-05-31.md`.